INCH-POUND

MS9483C
w/AMENDMENT 2
4 May 2017
SUPERSEDING
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W/AMENDMENT 1
14 February 2012

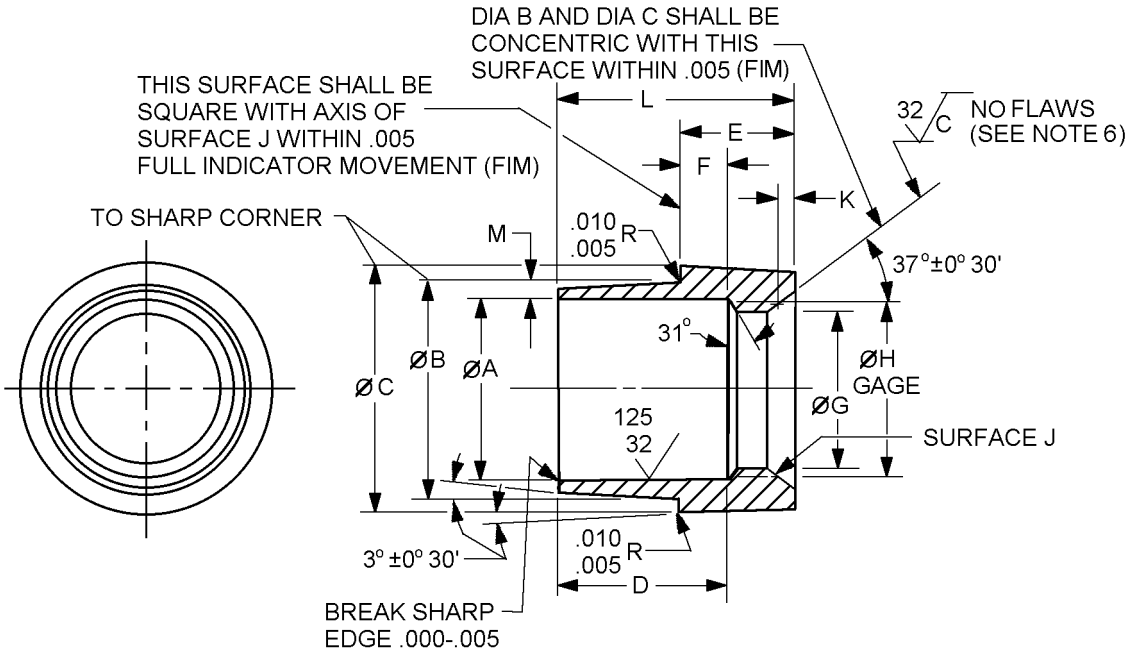
DETAIL SPECIFICATION SHEET

FERRULE, BRAZING, TUBE FITTING - CRES SAE-AMS5646, STRAIGHT

Inactive for new design after 20 September 2000.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and SAE-AS4841.



Inches	mm
.005	0.13
.010	.025

See notes at end of figure.

FIGURE 1. Ferrule, brazing, tube fitting straight.

AMSC N/A

FSC 4730

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Dash number	Tube OD (reference) inches (mm)	A diameter +.002 (0.05) -.000 inches (mm)	B diameter +.000 -.003 (0.08) inches (mm)	C diameter +.000 -.003 (0.08) inches (mm)	D inches (mm)
-02	.125 (3.18)	.115 (2.92)	.172 (4.37)	.267 (6.78)	.346 (8.79)
-03	.188 (4.78)	.178 (4.52)	.234 (5.94)	.329 (8.36)	.358 (9.09)
-04	.250 (6.35)	.240 (6.10)	.297 (7.54)	.383 (9.73)	.376 (9.55)
-05	.313 (7.95)	.302 (7.67)	.366 (9.30)	.445 (11.30)	.376 (9.55)
-06	.375 (9.53)	.365 (9.27)	.432 (10.97)	.502 (12.75)	.376 (9.55)
-07	.438 (11.13)	.428 (10.87)	.494 (12.55)	.564 (14.33)	.403 (10.24)
-08	.500 (12.70)	.489 (12.42)	.562 (14.27)	.682 (17.32)	.432 (10.97)
-09	.562 (14.27)	.551 (14.00)	.626 (15.90)	.743 (18.87)	.451 (11.46)
-10	.625 (15.88)	.614 (15.60)	.690 (17.53)	.797 (20.24)	.479 (12.17)
-11	.688 (17.48)	.677 (17.20)	.762 (19.35)	.908 (23.06)	.510 (12.95)
-12	.750 (19.05)	.739 (18.77)	.826 (20.98)	.972 (24.69)	.510 (12.95)
-14	.875 (22.23)	.864 (21.95)	.953 (24.21)	1.097 (27.86)	.510 (12.95)
-16	1.000 (25.40)	.989 (25.12)	1.081 (27.46)	1.222 (31.04)	.510 (12.95)
-18	1.125 (28.58)	1.114 (28.30)	1.209 (30.71)	1.409 (35.79)	.510 (12.95)
-20	1.250 (31.75)	1.239 (31.47)	1.339 (34.01)	1.534 (38.96)	.510 (12.95)
-24	1.500 (38.10)	1.484 (37.69)	1.609 (40.87)	1.784 (45.31)	.510 (12.95)
-28	1.750 (44.45)	1.734 (44.04)	1.882 (47.80)	2.159 (54.84)	.510 (12.95)
-32	2.000 (50.80)	1.984 (50.39)	2.159 (54.84)	2.409 (61.19)	.510 (12.95)
-40	2.500 (63.50)	2.484 (63.09)	2.659 (67.54)	2.911 (73.94)	.510 (12.95)
-48	3.000 (76.20)	2.984 (75.79)	3.172 (80.57)	3.411 (86.64)	.510 (12.95)

See notes at end of figure.

FIGURE 1. Ferrule, brazing, tube fitting straight - Continued.

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Dash number	E inches (mm)	F inches (mm)	G diameter inches (mm)	H diameter gage inches (mm)	K inches (mm)
-02	.235 (5.97)	.100 (2.54)	.090 (2.29)	.124 (3.15)	.041 (1.04)
-03	.235 (5.97)	.100 (2.54)	.150 (3.81)	.185 (4.70)	.040 (1.02)
-04	.235 (5.97)	.100 (2.54)	.200 (5.08)	.242 (6.15)	.036 (0.91)
-05	.235 (5.97)	.100 (2.54)	.260 (6.60)	.302 (7.67)	.035 (0.89)
-06	.228 (5.79)	.100 (2.54)	.320 (8.13)	.358 (9.09)	.031 (0.79)
-07	.255 (6.48)	.120 (3.05)	.385 (9.78)	.430 (10.92)	.036 (0.91)
-08	.285 (7.24)	.120 (3.05)	.438 (11.13)	.504 (12.80)	.044 (1.12)
-09	.295 (7.49)	.150 (3.81)	.500 (12.70)	.555 (14.10)	.036 (0.91)
-10	.302 (7.67)	.150 (3.81)	.562 (14.27)	.621 (15.77)	.039 (0.99)
-11	.305 (7.75)	.120 (3.05)	.625 (15.88)	.698 (17.73)	.048 (1.22)
-12	.306 (7.77)	.120 (3.05)	.688 (17.48)	.769 (19.53)	.054 (1.37)
-14	.325 (8.26)	.130 (3.30)	.812 (20.62)	.894 (22.71)	.054 (1.37)
-16	.342 (8.69)	.140 (3.56)	.938 (23.83)	1.021 (25.93)	.055 (1.40)
-18	.370 (9.40)	.140 (3.56)	1.062 (26.97)	1.174 (29.82)	.074 (1.88)
-20	.397 (10.08)	.160 (4.06)	1.188 (30.18)	1.299 (32.99)	.074 (1.88)
-24	.405 (10.29)	.180 (4.57)	1.438 (36.53)	1.532 (38.91)	.062 (1.57)
-28	.405 (10.29)	.100 (2.54)	1.688 (42.88)	1.839 (46.71)	.100 (2.54)
-32	.405 (10.29)	.100 (2.54)	1.938 (49.23)	2.099 (53.31)	.107 (2.72)
-40	.405 (10.29)	.100 (2.54)	2.438 (61.93)	2.594 (65.89)	.117 (2.97)
-48	.405 (10.29)	.100 (2.54)	2.938 (74.63)	3.099 (78.71)	.107 (2.72)

See notes at end of figure.

FIGURE 1. Ferrule, brazing, tube fitting straight - Continued.

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Dash number	L reference inches (mm)	M minimum inches (mm)	Approx. weight lb/100 (kg/100)
-02	.481 (12.22)	.023 (.58)	.356 (0.16)
-03	.493 (12.52)	.023 (.58)	.488 (0.22)
-04	.511 (12.98)	.023 (.58)	.615 (0.28)
-05	.511 (12.98)	.027 (.69)	.783 (0.36)
-06	.504 (12.80)	.028 (.71)	.894 (0.41)
-07	.538 (13.67)	.028 (.71)	1.090 (0.49)
-08	.597 (15.16)	.031 (.79)	1.910 (0.87)
-09	.596 (15.14)	.032 (.81)	2.140 (0.97)
-10	.631 (16.03)	.033 (.84)	2.340 (1.06)
-11	.695 (17.65)	.037 (.94)	3.350 (1.52)
-12	.696 (17.68)	.038 (.97)	3.670 (1.66)
-14	.705 (17.91)	.039 (.99)	4.430 (2.01)
-16	.712 (18.08)	.041 (1.04)	5.220 (2.37)
-18	.740 (18.80)	.042 (1.07)	7.410 (3.36)
-20	.747 (18.97)	.045 (1.14)	6.670 (3.03)
-24	.735 (18.67)	.057 (1.45)	11.020 (5.00)
-28	.815 (20.70)	.069 (1.75)	18.360 (8.33)
-32	.815 (20.70)	.082 (2.08)	21.840 (9.91)
-40	.815 (20.70)	.082 (2.08)	29.900 (13.56)
-48	.815 (20.70)	.089 (2.26)	32.920 (14.93

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified tolerances for linear dimensions are ±.010 inch (0.25 mm) and angular dimensions are ±5°, all diameters concentric within .010 inch (0.25 mm) FIM.
4. Unless otherwise specified machined surfaces to be 125µ inches (3.18µm) in accordance with ASME B46.1
5. Unless otherwise specified break sharp edges .003 - .015 inches (0.08 - 0.38 mm).
6. Surface J, conical sealing surface, shall be free, of flaws, chatter marks, and low spots when viewed with the unaided eye.

FIGURE 1. Ferrule, brazing, tube fitting straight - Continued.

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REQUIREMENTS:

Dimensions and configurations: See figure 1.

Material: Steel, corrosion and heat-resistant in accordance with SAE-AMS5646.

Max operating temperature: Oxidation resistance up to 1500°F (816°C).

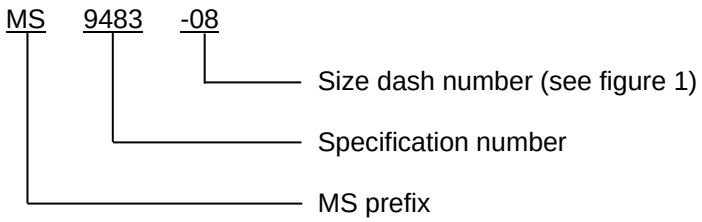
Max operating pressure: 1500 psi (10 MPa).

Finishing: Optional or abrasive finishing by silicon carbide only.

All parts shall be fluorescent penetrant inspected in accordance with ASTM E1417/E1417M.

Fluorescent penetrant shall be in accordance with SAE-AMS2644.

Part or Identifying Number (PIN): The PIN consists of the MS prefix, specification number, and a dash size number. Do not use unassigned PINs.



PIN Example:

MS9483-08 identifies a ferrule, brazing straight, tube .500 inch (tube), CRES.

Mark PIN and manufacturer's identification on package.

Intended usage: Jet engine, miscellaneous parts, oil, fuel, and de-icing kits.

Not to be used on hydraulic, oxygen or applications requiring pressures of 3000 psi (21 MPa).

Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Referenced documents shall be of the issue in effect on date of invitations for bid.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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Referenced documents. In addition to SAE-AS4841, this document references the following:

ASME B46.1
ASTM E1417/E1417M
SAE-AMS2644
SAE-AMS5646

CONCLUDING MATERIAL

Custodians:
Navy - AS
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2017-056)

Review activity:
Navy - SA

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